

● MEDIUM

● ALGEBRA

Linear functions

03

10 questions · ~15 min

Q1

GRID-IN ALGEBRA

$$c(x) = mx + 500$$

A company's total cost $c(x)$, in dollars, to produce x shirts is given by the function above, where m is a constant and $x > 0$. The total cost to produce 100 shirts is \$800. What is the total cost, in dollars, to produce 1000 shirts? (Disregard the \$ sign when gridding your answer.)

STUDENT-PRODUCED RESPONSE

○	/○	/○	○
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The relationship between two variables, x and y , is linear. For every increase in the value of x by 1, the value of y increases by 8. When the value of x is 2, the value of y is 18. Which equation represents this relationship?

A. $y = 2x + 18$

B. $y = 2x + 8$

C. $y = 8x + 2$

D. $y = 3x + 26$

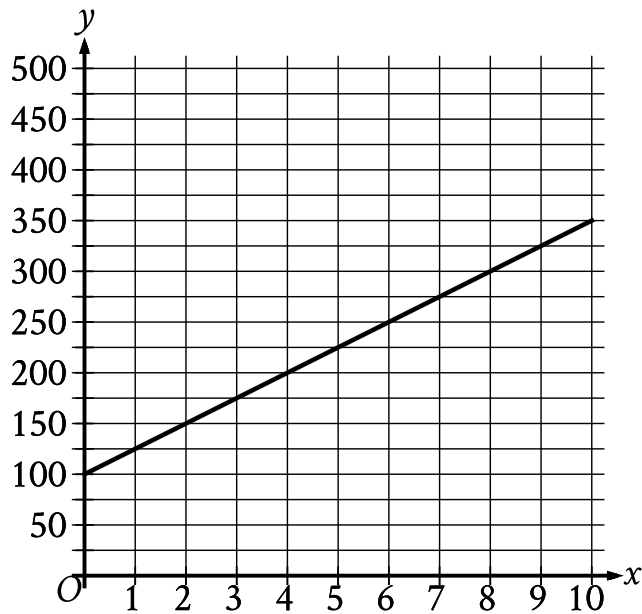
The function h is defined by $hx = 4x + 28$. The graph of $y = hx$ in the xy -plane has an x -intercept at $a, 0$ and a y -intercept at $0, b$, where a and b are constants. What is the value of $a + b$?

- A. 21
- B. 28
- C. 32
- D. 35

$$w(t) = 300 - 4t$$

The function w models the volume of liquid, in milliliters, in a container t seconds after it begins draining from a hole at the bottom. According to the model, what is the predicted volume, in milliliters, draining from the container each second?

- A. 300
- B. 296
- C. 75
- D. 4



The graph of the function f , where $y = f(x)$, gives the total cost y , in dollars, for a certain video game system and x games. What is the best interpretation of the slope of the graph in this context?

- A. Each game costs \$25.
- B. The video game system costs \$100.
- C. The video game system costs \$25.
- D. Each game costs \$100.

A team of workers has been moving cargo off of a ship. The equation below models the approximate number of tons of cargo, y , that remains to be moved x hours after the team started working.

$$y = 120 - 25x$$

The graph of this equation in the xy -plane is a line. What is the best interpretation of the x -intercept in this context?

- A. The team will have moved all the cargo in about 4.8 hours.
- B. The team has been moving about 4.8 tons of cargo per hour.
- C. The team has been moving about 25 tons of cargo per hour.
- D. The team started with 120 tons of cargo to move.

If $f(x) = x + 6$ and $g(x) = 6x$, what is the value of $7f(2) - g(2)$?

A. -4

B. 8

C. 42

D. 44

For the linear function h , the graph of $y = hx$ in the xy -plane passes through the points $(7, 21)$ and $(9, 25)$. Which equation defines h ?

A. $hx = \frac{1}{2}x - \frac{7}{2}$

B. $hx = 2x + 7$

C. $hx = 7x + 21$

D. $hx = 9x + 25$

The table gives the number of hours, h , of labor and a plumber's total charge fh , in dollars, for two different jobs.

h	fh
1	155
3	285

There is a linear relationship between h and fh . Which equation represents this relationship?

- A. $fh = 25h + 130$
- B. $fh = 130h + 25$
- C. $fh = 65h + 90$
- D. $fh = 90h + 65$

In the xy -plane, the graph of the linear function f contains the points 0, 3 and 7, 31.
Which equation defines f , where $y = fx$?

A. $fx = 28x + 34$

B. $fx = 3x + 38$

C. $fx = 4x + 3$

D. $fx = 7x + 3$